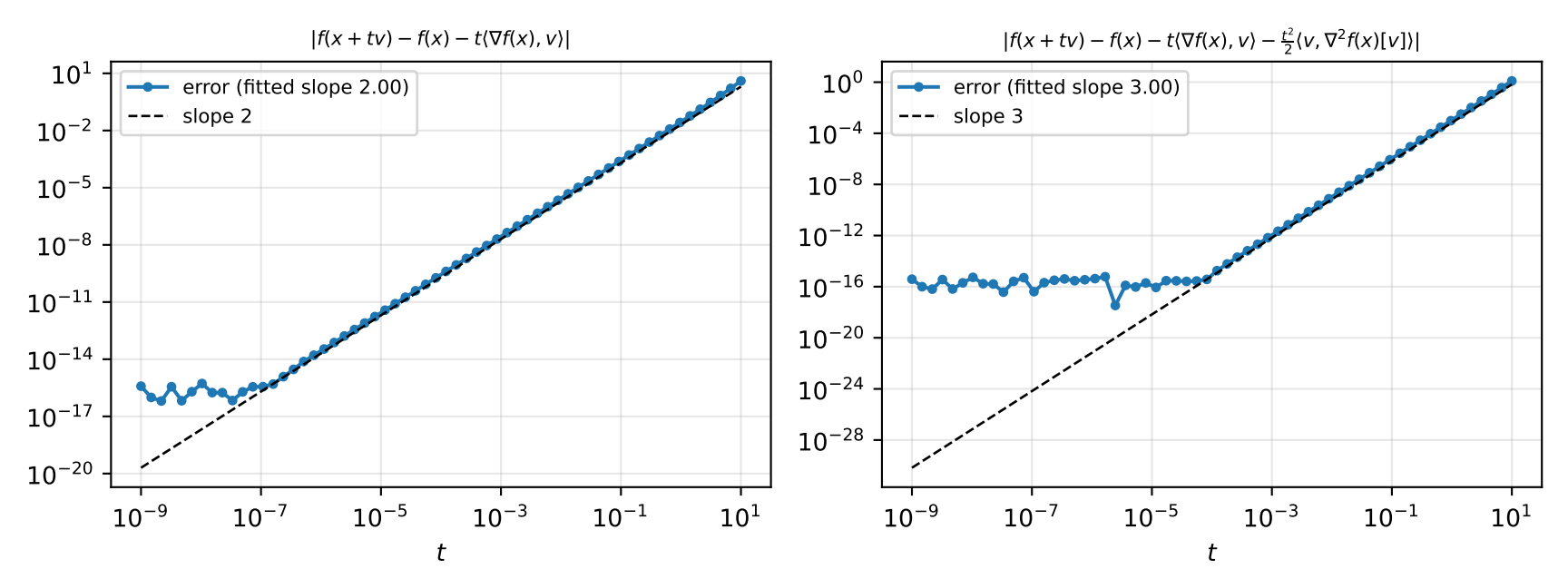
$|f(x+tv) - f(x) - t\langle \nabla f(x), v\rangle|$
error (fitted slope 2.00)
slope 2
10^{1}
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10^{-14}
10^{-17}
10^{-20}
10^{-9}
10^{-7}
10^{-5}
10^{-3}
10^{-1}
10^{1}
t
$|f(x+tv) - f(x) - t\langle \nabla f(x), v\rangle - \frac{t^2}{2}\langle v, \nabla^2 f(x)[v]\rangle|$
error (fitted slope 3.00)
slope 3
10^{0}
10^{-4}
10^{-8}
10^{-12}
10^{-16}
10^{-20}
10^{-24}
10^{-28}
10^{-9}
10^{-7}
10^{-5}
10^{-3}
10^{-1}
10^{1}
t